

Neil Artus

(+63) 9611249491 | artus.neil.c@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | General Trias, Cavite, Philippines

TECHNICAL SKILLS

Languages: Python, Go, C#, JavaScript, TypeScript, SQL

Frameworks: FastAPI, Flask, React, Next.js, Vite, Tailwind CSS, .NET, Fiber, LangChain, LangGraph, Semantic Kernel, PyTorch, TensorFlow

Developer Tools: Azure DevOps, Docker, Firebase, Hetzner, Nginx, Gunicorn, Ubuntu Linux, Windows Server 2022, CI/CD, Microsoft Entra ID (Azure AD), Okta, SonarQube, Black Duck, Coverity, Cloudflare, Git, ESP32, Raspberry Pi

Libraries: OpenTelemetry, Cosmos DB, Qdrant, Microsoft SQL Server, Entity Framework, SQLite, Pandas, Pydantic, Azure OpenAI, Azure AI Foundry, Google Vertex AI, Radix UI, TanStack Table, Recharts, Flask-SocketIO, Gemini API, MobileNet

EXPERIENCE

AI Engineer Intern

Mar 2026 – Jul 2026

Emerson

Metro Manila, Philippines

- Architected a multi-agent cybersecurity automation platform in C# using Semantic Kernel and custom Model Context Protocol (MCP via stdio transport) and REST integrations using tool registries with strongly typed C# record schemas, saving 16,000+ hours and \$90,000+ in operational expenses
- Engineered an agent telemetry monitoring service in ASP.NET Core, achieving sub-100ms p99 query latency through Entity Framework B-tree index tuning, connection pooling, denormalized read-optimized schemas and in-memory caching middleware
- Refactored and modernized (C# 12, extensive use of primary constructors, collection expressions, LINQ, strongly-typed immutable Data Transfer Objects) legacy DeltaV compliance software to .NET 8, consolidating multi-file CIS (Center for Internet Security) Excel benchmarks into an aggregated single-source CSV buffer with asynchronous file streaming and strict type checking to accelerate report generation by 30%
- Built Azure DevOps CI/CD pipelines integrating SonarQube static analysis, Black Duck SCA, and Coverity SAST quality gates with automated build signing and artifact publishing, cutting build and deploy cycle times to IIS on Windows Enterprise Server 2022 by 85%
- Modernized compliance UI by replacing legacy Material UI components with Radix UI primitives and refactoring nested useEffect pipelines into local useState hooks, reducing bundle size by 20% with dynamic code splitting
- Hardened internal microservice endpoints by implementing ASP.NET Core authentication middleware to validate Okta and Microsoft Entra ID Bearer tokens, enforcing RBAC claim extraction and tenant access boundaries across 100+ internal accounts

AI Engineer Intern (GenAI Research)

Jun 2025 – Aug 2025

JG Summit Holdings Inc.

Metro Manila, Philippines

- Architected an Agentic RAG service in Flask and React using Azure OpenAI (GPT-4) and LangChain tool registries with autonomous routing across 12+ financial data endpoints, cutting manual corporate strategy and treasury analysis time by 70%
- Built an asynchronous document ingestion pipeline using semantic chunking, metadata filtering, and hybrid search over financial PDFs, generating SHA-256 content hashes and source attribution metadata in SQLite to eliminate hallucinations, provide citations, and traceability
- Engineered automated data validation pipelines with Pydantic and SQL, enforcing strict data types and string-length constraints on LLM JSON outputs to achieve 93% validation accuracy prior to database persistence
- Standardized modular service templates and internal API documentation, reducing new engineer onboarding ramp-up time by 40%

PROJECTS

AGOS: Agentic Philippine Market Intelligence Platform

- Architected a concurrent multi-agent runtime in FastAPI using asyncio TaskGroups to orchestrate specialized workers (Research, Portfolio, Risk) and stream DeepSeek token deltas via Server-Sent Events
- Engineered a decoupled market scraping pipeline using curl_cffi TLS fingerprinting and across PSE EDGE (280+ securities), PSA OpenStat, and BSP REST APIs, processing 350+ daily metrics with SHA-256 natural key deduplication into Azure Cosmos DB

- Hardened API services with SlowAPI rate limiting , Google OAuth (Firebase) JWT validation via FastAPI dependency injection, and parameterized Cosmos DB queries

Enterprise HR Automation Engine | Independent Project

- Developed a Python-based automation engine using Celery and Redis task queues to process 10,000+ employee records, reducing manual reporting latency by 50% per pay cycle
- Orchestrated backend services on Hetzner Linux servers, maintaining 99.9% uptime via Nginx load balancing and automated systemd service management
- Implemented secure authentication and Role-Based Access Control (RBAC) using Microsoft Entra ID to safeguard sensitive personal records

MediSeen: Clinical Decision Support Platform

- Developed an AI clinical decision support progressive web application using Gemini Flash models, earning 1st Place at the International Conference on Artificial Intelligence (ICAI 2026)
- Achieved 400ms time-to-first-token (TTFT) by streaming model reasoning steps via Server-Sent Events (SSE) and optimizing client-side image compression using non-blocking asyncio patterns
- Deployed Azure App Services backend integrated with Cosmos DB, optimizing partition key distribution to achieve 98ms average database query latency under concurrent load
- Configured Nginx Reverse Proxy, Cloudflare DNS caching and SSL/TLS termination, designing a responsive Tailwind CSS interface that raised measured System Usability Scale (SUS) scores by 15%

CNN-Osteo: Edge Thermal Diagnostic Classifier

- Trained a MobileNetV3-Small architecture (~60M FLOPs) on 2,236 medical thermal images, applying post-training INT8 quantization to compress model size to 4.3 MB for edge deployment
- Achieved 99.40% validation and 98.19% test accuracy; deployed inference scripts to a Raspberry Pi 5 interfaced with a thermal camera module for real-time diagnostic grading
- Won 1st Place at Cavite Innovex 2025 Colloquium (COESCA Research of the Year) for edge computer vision applied to osteoarthritis detection

Alersense: Low-Latency IoT Telemetry Monitor

- Engineered ESP32 firmware in C++ using a FreeRTOS queue to decouple hardware sensor polling from network transmission, enforcing a non-blocking drop-on-full policy to prevent memory exhaustion
- Eliminated dynamic memory allocation via pass-by-value struct copies, streaming heart rate, SpO2, and temperature metrics at fixed 5-second intervals to Flask-SocketIO with sub-50ms end-to-end latency

Deterministic Finite Automata (DFA) Compiler

- Engineered an $O(n)$ time DFA execution engine in Python using hash-map transition adjacency tables, typed error markers, and early-exit trap state validation
- Implemented compilation modules transforming DFAs into stack-tracing PDAs and right-linear CFGs, utilizing breadth-first search with signature-based visited-set pruning to resolve nondeterministic ϵ -transitions
- Built a React/SVG visualization engine rendering polar layout geometry and Bézier curves, verified via table-driven pytest suites asserting cross-model DFA/PDA/CFG acceptance parity

EDUCATION

De La Salle University - Dasmariñas

Dasmariñas, Cavite, Philippines

Bachelor of Science in Computer Science with Specialization in Intelligent Systems (AI)

- Magna Cum Laude, GPA: 3.52/4.0, TOEIC: 955/990
- Leadership: Secretary & Codeyssey Program Head for the Computer Science Program Council; managed technical workshops and hackathons for 1,000+ students
- Honors: CSPC Leadership Award Recipient, Dean's Honor List
- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Web Development, Computer Networks, Operating Systems, Software Engineering